# Functional annotation

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Jacques Dainat Ph.D.

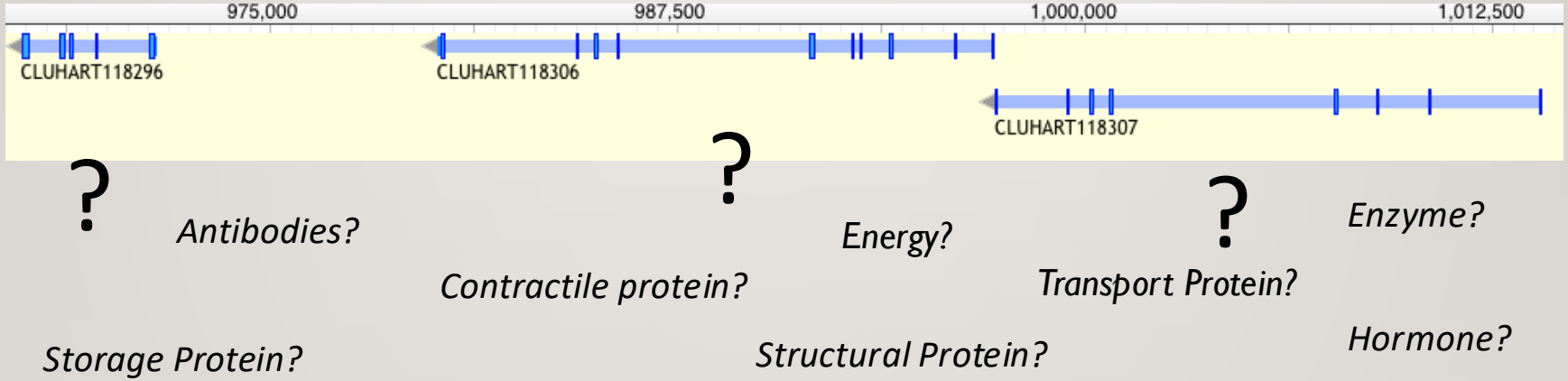


# Functional annotation – why?

Understanding the function of gene product is key to understanding how a limited number of interacting gene products can generate life, from simple unicellular organisms to the incredibly complex multi-cellular Homo sapiens.

Rison,S.C., Hodgman,T.C. and Thornton,J.M. (2000) Comparison of functional annotation schemes for genomes. Funct. Integr. Genomics, 1, 56–69.

Proteins vary in structure as well as function



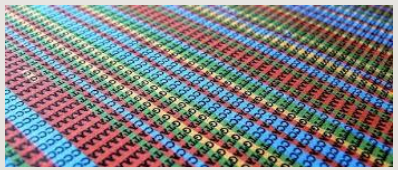
# Functional annotation – how?

- Experimentally (accurate)
  - Mutants, knockout, etc.

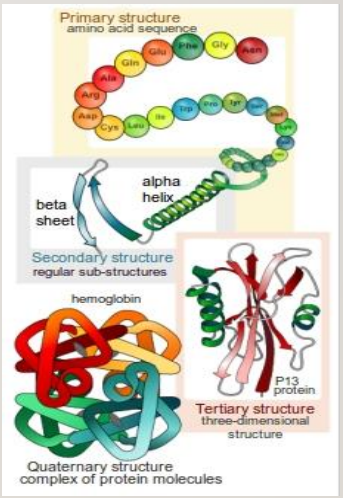


Mice homozygous for the diabetes 3J spontaneous mutation

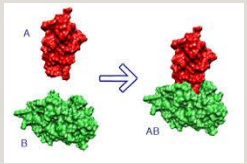
- Computationally (limited accuracy)
  - Sequence-based



  - Structure based




  - Protein-protein interaction



# Methods - Sequence-based

- Based on similarity

  => Best blast hit

```
Q GLMDTAFEHIKATGGLTTESNYPYKGEDATCNS-K
  GLM+ AFE+IK +GG+TTES YPY+  + TC++ +
S GLMENAFEYIKHSGGITTESAYPYRAANGTCDAVR
```

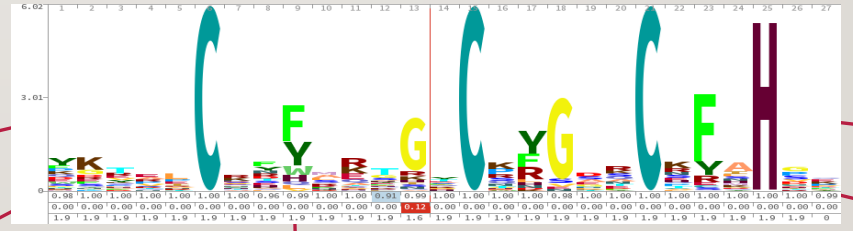
- Based on Motif

  => Proscan, MEME, QuasiMotiFinder

**D-X-[KR]-P-{WYF}-X5**

- Based on Profile

  => HMM or other statistical signature



Whole sequence
e.g. Psi-BLAST*, PIRSF

domain
e.g. PFAM

Localization (e.g membrane, golgi, secreted)
e.g. SignalP, TMHMM

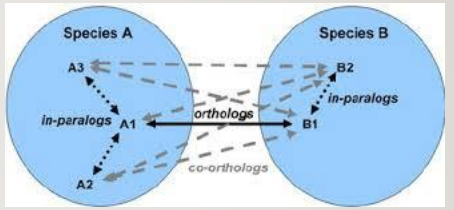
structural classification
e.g. SUPERFAMILY

...

# Use of evolutionary relationship (Orthology)

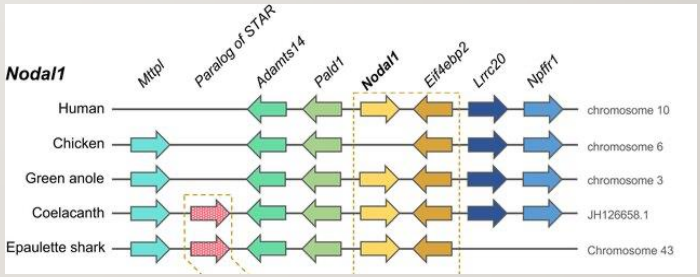
- Based on Clustering

  => KOG (Eukaryotic) / COG(Prokaryotic)



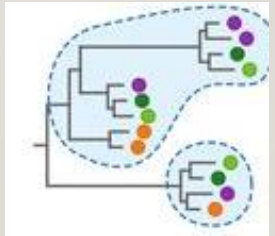

- Based on synteny

  => conservation of gene order/organization across species




- Based on phylogeny

  => Quite complicated at large scale



# Methods - Structure-based

Focuses on the **3D conformation of a single protein** and similarity to know structures to infer function.



- Global structure

  Classification of overall protein folds and domain (e.g. SCOP, CATH, FSSP)

- local structure

  Functionally relevant *parts of the structure* e.g. clefts, ligand-binding pockets, **interfaces**, surfaces.

  It often suggests where ligands, cofactors, or partner proteins may interact (e.g. CASTp, fpocket, ProFunc).

- active-site residues (atomic level)

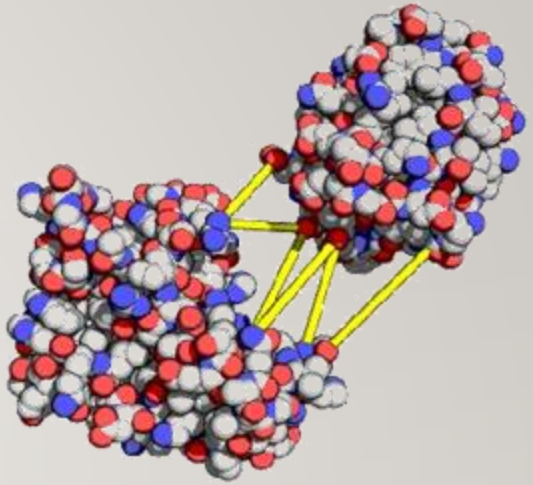
  Specific catalytic or ligand-binding residues. Residues are often more conserved than the overall fold (e.g. PDBSiteScan, Catalytic Site Atlas)

# Methods – Protein-Protein interaction

- Structure-guided interaction prediction (who interacts)

  Docking and modeling of protein complexes (e.g. HADDOCK, ClusPro)

  Identifies potential interaction partners and functional interfaces.

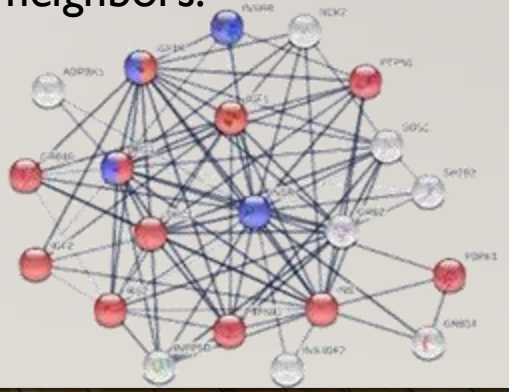


- Pathway and network context (what role)

  Integration of predicted PPIs into biological networks (e.g. STRING).

  Guilt-by-association: uncharacterized proteins may inherit function from well-annotated neighbors.

  Supports placement of proteins into pathways and complexes.



# Functional annotation is actually kind of complex…

- Multidimensional problem:

  A protein can have a molecular function, a cellular role, and be part of a functional complex or pathway

- Descriptive levels

  Molecular function can be illustrated at multiple levels e.g. '**enzyme**' category versus a more specific '**protease**' assignment

- Similarities (structural or in sequence)  function.

  - Similar sequences but different functions (new domain => new combination => different function)

  - Different sequences may have same function (convergence) : Profiles helpful

  - Two proteins may have a similar fold but different functions

- Looks for conserved domains more reliable than whole sequence ?

  How to go from conserved domains to assigning a function to your protein?

=> Importance is to gather as much information as possible

# Sequence-based methods – WHY?

- The most used (popular)
- Quick
- Easy to use
- Accurate (>70%)
- Many resources: even structural domains information
- Less computationally demanding

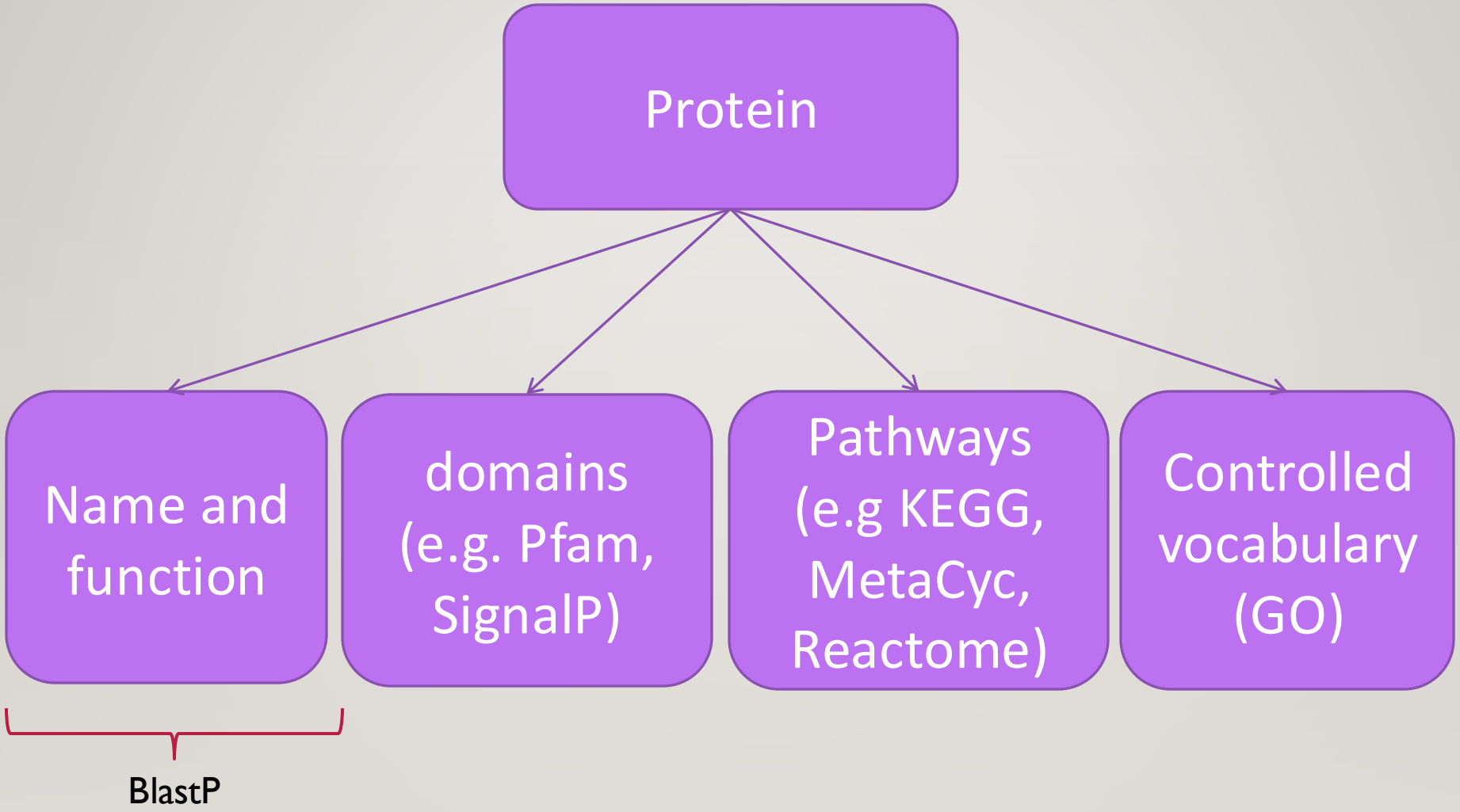
Watson JD, Sanderson S, Ezersky A, Savchenko A, Edwards A, Orengo C, Joachimiak A, Laskowski RA, Thornton JM: Towards fully automated structure-based function prediction in structural genomics: a case study. J Mol Biol. 2007, 367: 1511-1522. 10.1016/j.jmb.2007.01.063.

# Sequence-based methods – HOW?

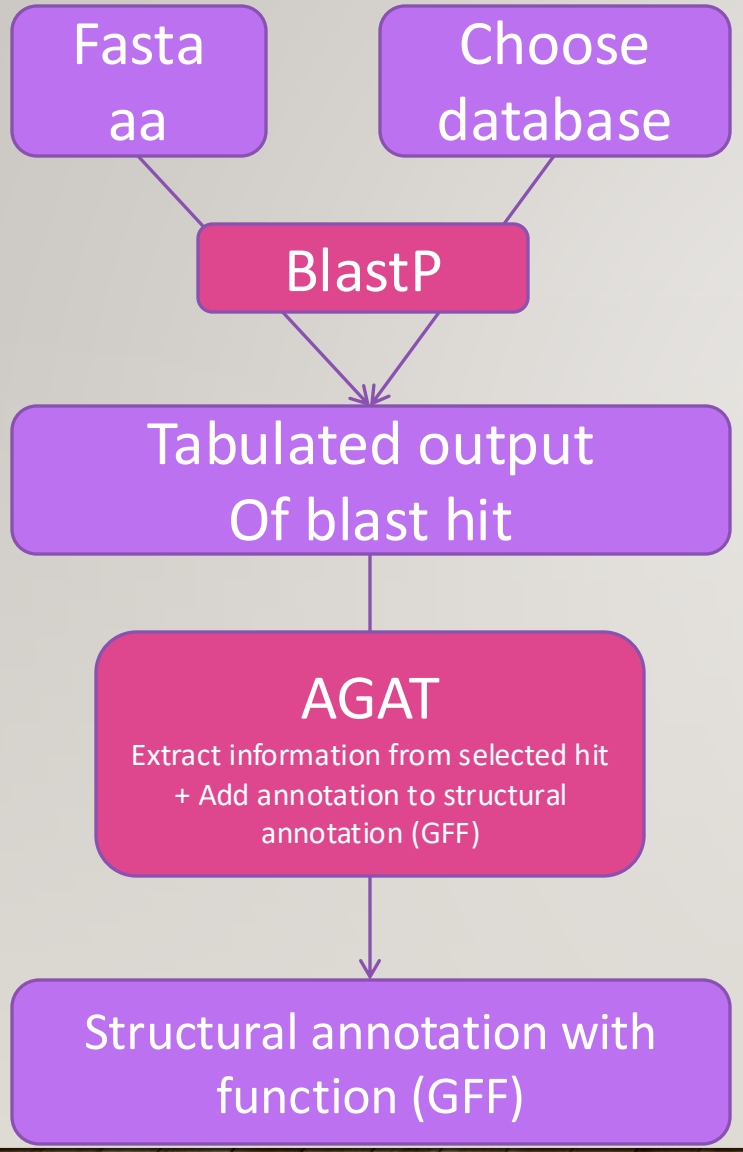
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# Sequence-based methods – HOW?



# Annotate the sequences functionally using Blast

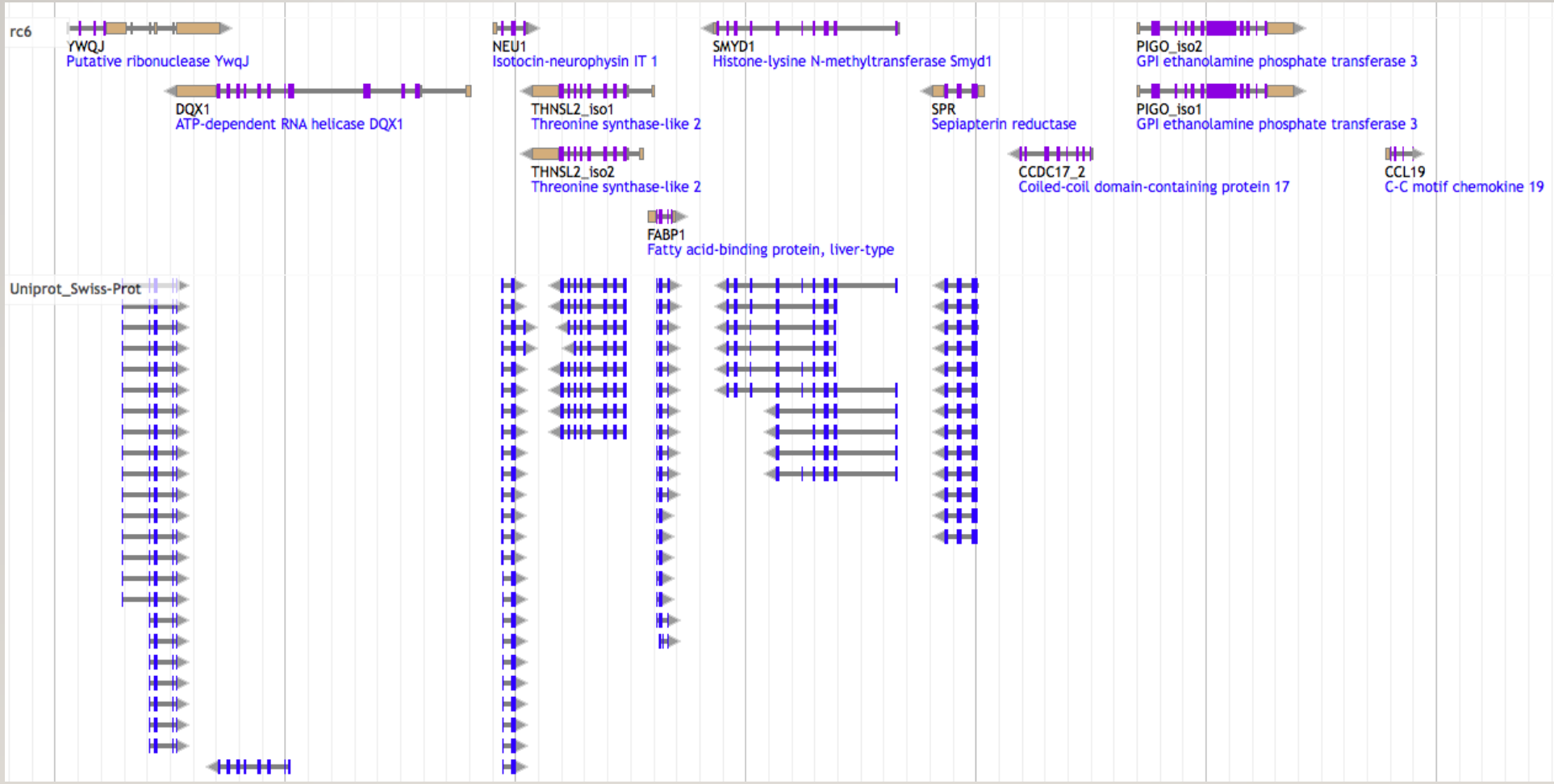



| Uniprot | Swissprot |
|---|---|
| exhaustive | reliable |

**How to filter ?**

- Minimum e-value
- Best blast hit or prioritized by species

# Annotate the sequences functionally using Blast



# Annotate the sequences functionally using Blast

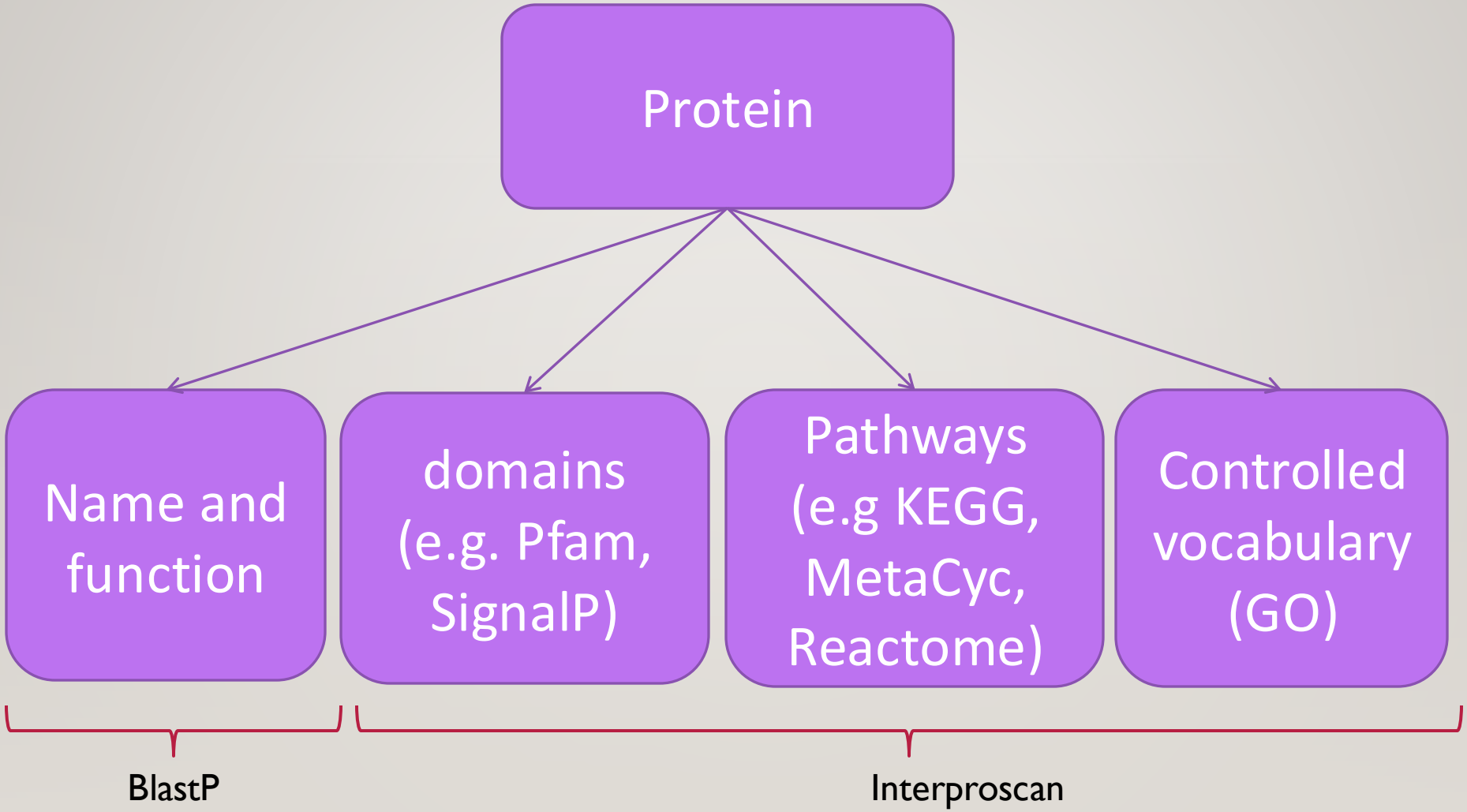
## Strengths

- Fairly fast and easy
- Allow gene naming (e.g. plip)
- Overall function (e.g. Phosphatidylglycerophosphatase and protein-tyrosine phosphatase 1)

## Limits

- Orthology not certain - best blast-hit does not equal orthologous!
- Bias due to well conserved domains
- Best Hit ( use as template) is not necessary the best annotated sequence to use => Could apply a prioritization rule (Human first, then mouse, etc).

# Sequence-based methods – HOW?



# Databases

| Database | Information | Comment |
|---|---|---|
| KEGG | Pathway | Kyoto Encyclopedia of Genes and Genomes |
| MetaCyc | Pathway | Curated database of experimentally elucidated metabolic pathways from all domains of life (NIH) |
| Reactome | Pathway | Curated and peer reviewed pathway database |
| GO | Gene Ontology | Three structured, controlled vocabularies (ontologies) : biological processes, cellular components and molecular functions |
| Pfam | Protein families | Multiple sequence alignments and hidden Markov models |
| **Interpro** | Protein families, domains and functional sites | Run separate search applications, and create a signature to search against Interpro. |
| Have a look on the Interpro web page: All the database they search into are listed. It gives a nice overview of different types of databases available. | | |

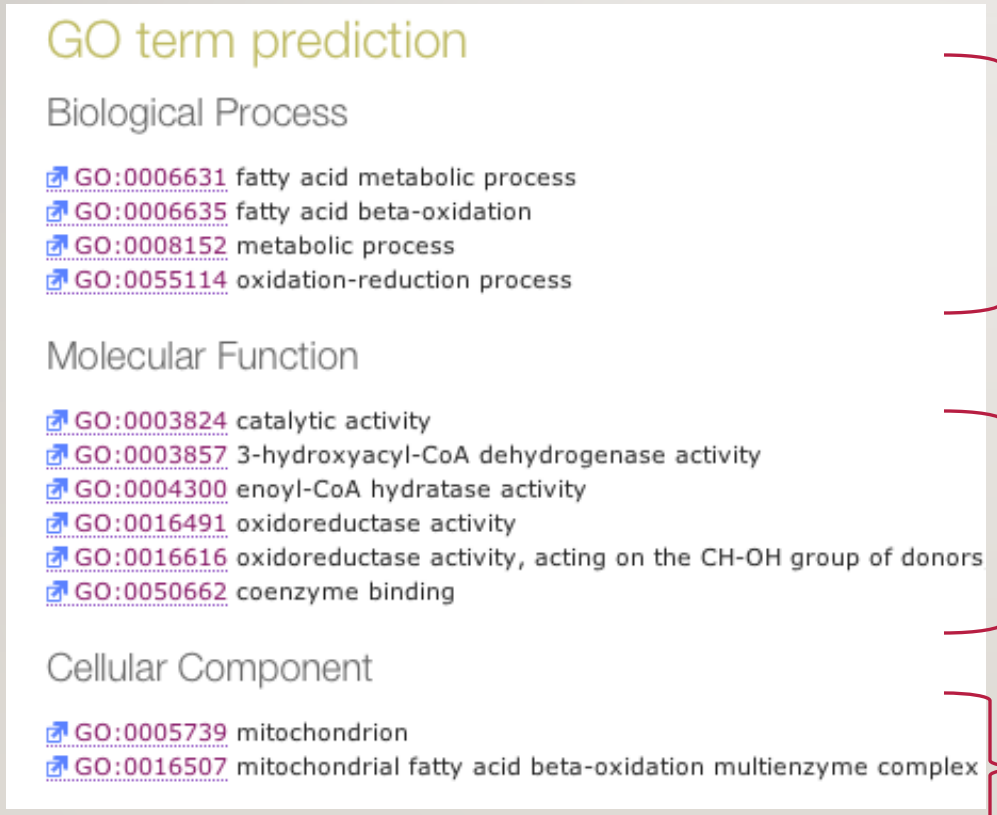
# Gene Ontology




**Gene Ontology (GO)** is a major bioinformatics initiative that uses a **controlled vocabulary** (> 60 000 terms) to standardize the representation of gene and gene product attributes across all species.

http://www.geneontology.org/

**Gene Ontology (GO)** organizes function into three complementary aspects:




pathways and larger processes made up of the activities of multiple gene products.

molecular activities of gene products

where gene products are active

# Tools

| Tool | Approach | Comment |
|---|---|---|
| Trinotate | Best blast hit + protein domain identification (HMMER/PFAM) + protein signal peptide and transmembrane domain prediction (signalP/tmHMM), and leveraging various annotation databases (eggNOG/GO/Kegg databases). | Partially automated |
| Annocript | Best blast hit | Collects the best-hit and related annotations (proteins, domains, GO terms, Enzymes, pathways, short) |
| Annot8r | Best blast hits | A tool for Gene Ontology, KEGG biochemical pathways and Enzyme Commission EC number annotation of nucleotide and peptide sequences. |
| Sma3s | Best blast hit + Best reciprocal blast hit + clusterisation | 3 annotation levels |
| afterParty | BLAST, InterProScan | web application |
| **Interproscan** | Run separate search applications<br>HMMs, fingerprints, patterns => InterPro | Created to unite secondary databases |
| Blast2Go | Best* blast hits<br>(picks out GO terms based on blast hits and uniprot with statistical significance test ) | Commercial !<br>Contain nice downstream analysis/visualization components<br>Plugin for Geneious or CLC bio Workbench |

- **InterPro**

  is an integrative database that combines the main **protein signature databases** (Pfam, PROSITE, SMART, CATH, SUPERFAMILY, etc.).

- **A signature**

  is a motif, a profile, or an HMM (Hidden Markov Model) that describes a protein family, a domain, an active site, or another functional region.

- **InterProScan**

  is the software that compares a given protein sequence against all these signature databases to assign potential functions.

# Interpro

## InterPro 106.0 • 19th June 2025

Features include:

- The addition of 683 InterPro entries.
- An update to Pfam [37.4].
- Integration of 727 new methods from the SMART (1), Pfam (613), NCBIFAM (78), SUPERFAMILY (1), PANTHER (4), CDD (30) databases.

### Contents and coverage

InterPro protein matches are now calculated for all UniProtKB and UniParc proteins. InterPro release 106.0 contains 48 679 entries (last entry: IPR058129), representing:

| | | |
|---|---|---|
| H | Homologous Superfamily | 3 512 |
| F | Family | 26 622 |
| D | Domain | 17 171 |
| R | Repeat | 385 |
| S | Site | |
| | ∟ Active Site | 133 |
| | ∟ Binding Site | 82 |
| | ∟ Conserved Site | 757 |
| | ∟ PTM | 17 |

## InterPro2GO

We have a total number of 30 207 GO terms mapped to InterPro entries.

List of InterPro2GO mappings. These are also available through the EBI GO browser QuickGO

### Member database information

| | Signature database | Version | Signatures | Integrated signatures |
|---|---|---|---|---|
| CATH | CATH-Gene3D | 4.3.0 | 6 631 | 2 823 (42.6%) |
| Conserved Domains | CDD | 3.21 | 19 902 | 4 948 (24.9%) |
| | HAMAP | 2025_01 | 2 391 | 2 386 (99.8%) |
| | NCBIFAM | 17.0 | 31 915 | 6 007 (18.8%) |
| PANTHER | PANTHER | 19.0 | 15 683 | 10 528 (67.1%) |
| Pfam | Pfam | 37.4 | 24 736 | 23 862 (96.5%) |
| PIR | PIRSF | 3.10 | 3 285 | 3 230 (98.3%) |
| PRINTS | PRINTS | 42.0 | 2 106 | 1 939 (92.1%) |
| prosite | PROSITE patterns | 2025_01 | 1 311 | 1 282 (97.8%) |
| prosite | PROSITE profiles | 2025_01 | 1 399 | 1 303 (93.1%) |
| SFLD | SFLD | 4 | 303 | 159 (52.5%) |
| SMART | SMART | 9.0 | 1 322 | 1 274 (96.4%) |
| | SUPERFAMILY | 1.75 | 2 019 | 1 645 (81.5%) |

pathway :

- MetaCyc
- Reactome

other licensed analysis available:

- Phobius (licensed software)
- SignalP
- SMART (licensed components)
- TMHMM

## Search against signature databases

InterProScan runs searches in parallel (Each database uses its own search engine:

- Pfam → HMMER (HMM profiles).
- PROSITE → regular expressions or profiles.
- SUPERFAMILY → structural HMMs.
- COILS → prediction of coiled-coil regions.

## Result aggregation

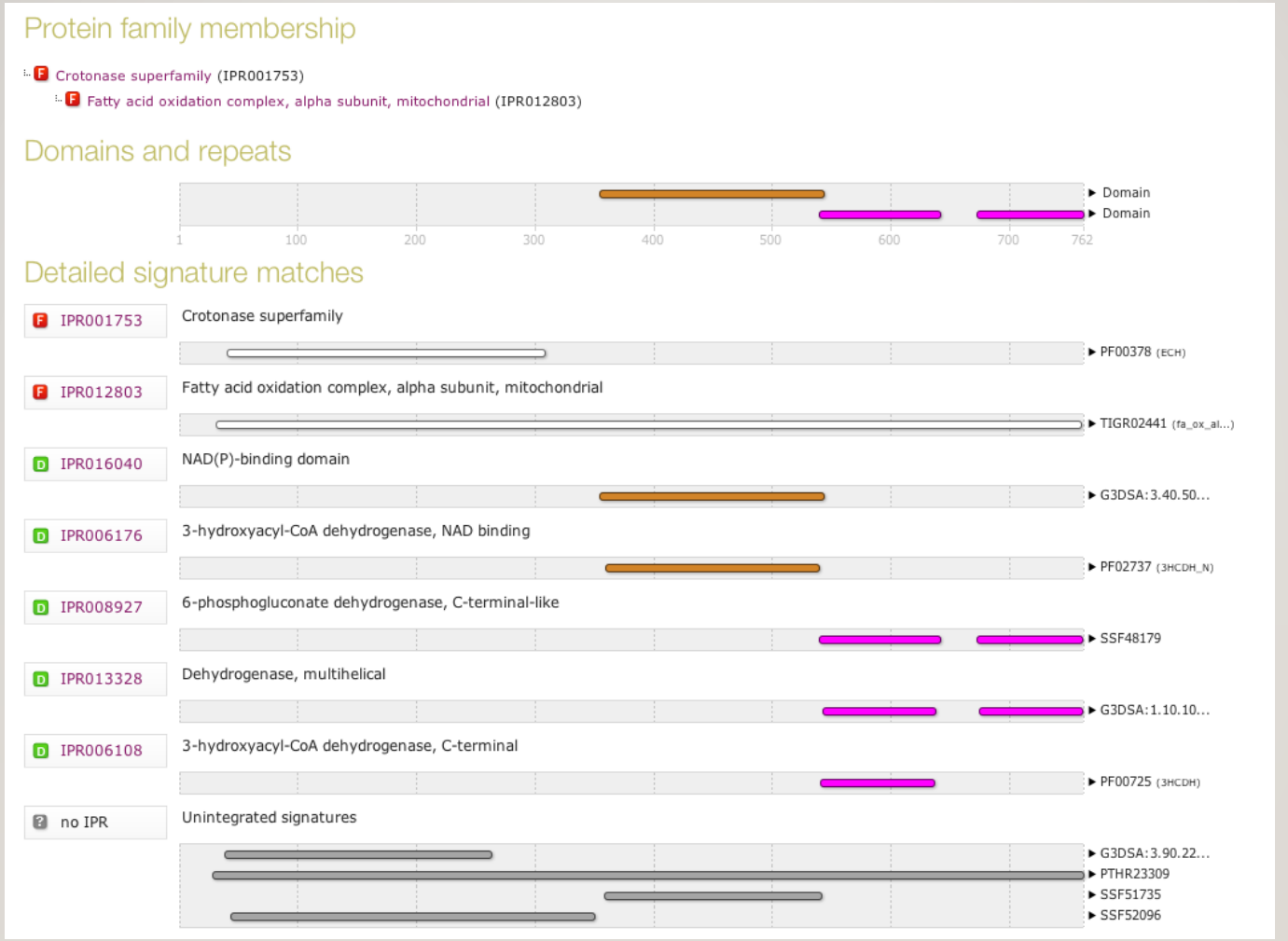
All matches are collected and for each match, InterProScan adds:

- The **InterPro ID** (if available, e.g. IPR002568).
- Associated **GO terms** (if --goterms is used).
- Associated **pathways** (if -pa is used).

## "iprlookup" option (default: enabled)

Links matches from member databases (e.g. Pfam: PF01623) to the corresponding **InterPro entry** (e.g. IPR002568).
This unifies results coming from different databases.

# Interproscan



# One word about network

Categorizations of gene function (e.g GO) in a hierarchy of categories is helpful

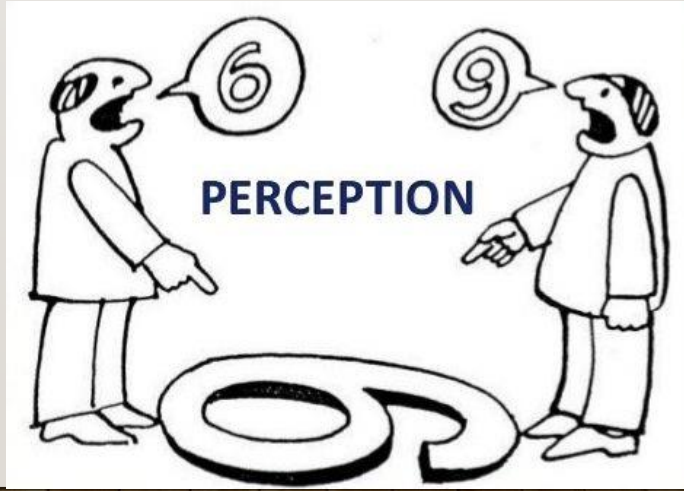
BUT

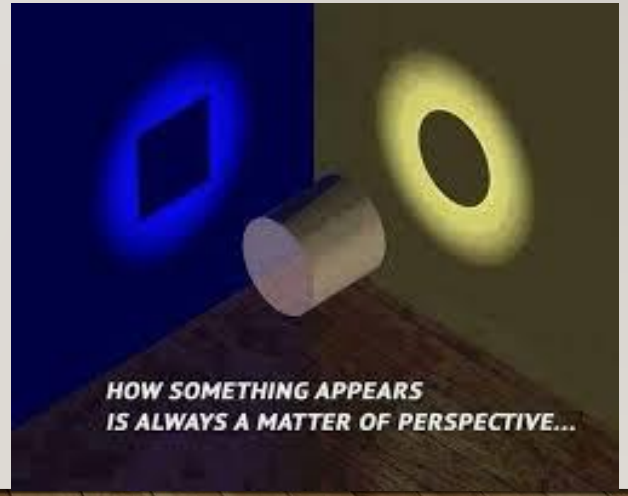
**gene has no function alone**

⇒Pathways / regulatory networks explain how genes interact so what they are doing!

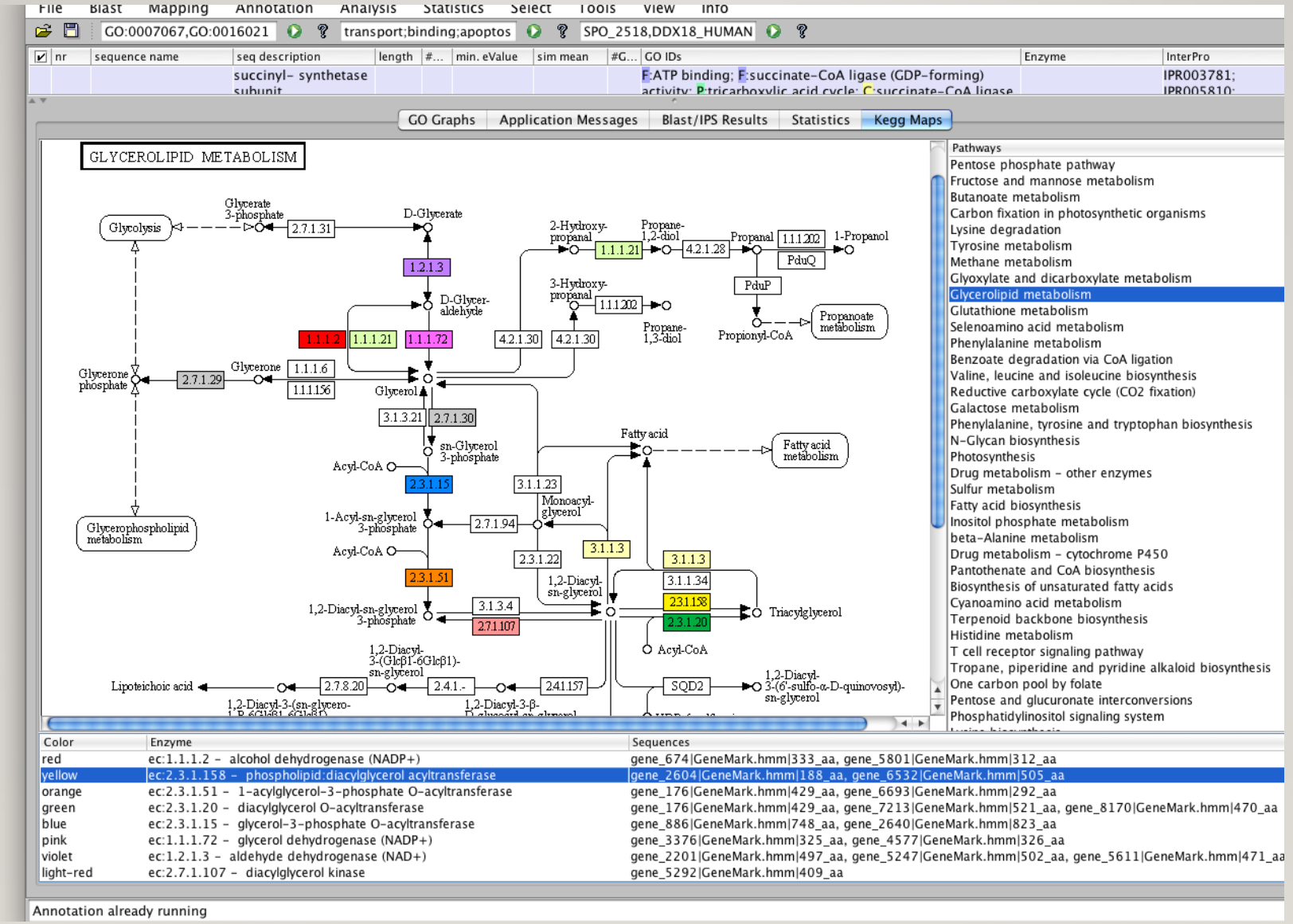
E.g. databases for pathway :

- KEGG
- MetaCyc
- Reactome
- UniPathway



# KEGG


File Blast Mapping Annotation Analysis Statistics Select Tools View Info
GO:0007067,GO:0016021
transport;binding;apoptos
SPO_2518,DDX18_HUMAN
nr
sequence name
seq description
length
min. eValue
sim mean
GO IDs
Enzyme
InterPro
succinyl- synthetase subunit
F:ATP binding; F:succinate-CoA ligase (GDP-forming)
IPR003781;
GO Graphs
Application Messages
Blast/IPS Results
Statistics
Kegg Maps
GLYCEROLIPID METABOLISM
Pathways
Pentose phosphate pathway
Fructose and mannose metabolism
Butanoate metabolism
Carbon fixation in photosynthetic organisms
Lysine degradation
Tyrosine metabolism
Methane metabolism
Glyoxylate and dicarboxylate metabolism
Glycerolipid metabolism
Glutathione metabolism
Selenoamino acid metabolism
Phenylalanine metabolism
Benzoate degradation via CoA ligation
Valine, leucine and isoleucine biosynthesis
Reductive carboxylate cycle (CO2 fixation)
Galactose metabolism
Phenylalanine, tyrosine and tryptophan biosynthesis
N-Glycan biosynthesis
Photosynthesis
Drug metabolism - other enzymes
Sulfur metabolism
Fatty acid biosynthesis
Inositol phosphate metabolism
beta-Alanine metabolism
Drug metabolism - cytochrome P450
Pantothenate and CoA biosynthesis
Biosynthesis of unsaturated fatty acids
Cyanoamino acid metabolism
Terpenoid backbone biosynthesis
Histidine metabolism
T cell receptor signaling pathway
Tropane, piperidine and pyridine alkaloid biosynthesis
One carbon pool by folate
Pentose and glucuronate interconversions
Phosphatidylinositol signaling system
Glycolysis
Glycerate 3-phosphate
2.7.1.31
D-Glycerate
1.2.1.3
D-Glycer-aldehyde
1.1.1.2
1.1.1.21
1.1.1.72
2-Hydroxy-propanal
Propane-1,2-diol
4.2.1.28
Propanal
1.1.1.202
1-Propanol
PduQ
PduP
3-Hydroxy-propanal
Propane-1,3-diol
Propionyl-CoA
Propanoate metabolism
4.2.1.30
Glycerone phosphate
2.7.1.29
Glycerone
1.1.1.6
1.1.1.156
Glycerol
3.1.3.21
2.7.1.30
sn-Glycerol 3-phosphate
Acyl-CoA
2.3.1.15
Fatty acid
Fatty acid metabolism
3.1.3.23
Monoacyl-glycerol
1-Acyl-sn-glycerol 3-phosphate
2.7.1.94
Glycerophospholipid metabolism
2.3.1.22
3.1.1.3
1,2-Diacyl-sn-glycerol
2.3.1.51
3.1.1.34
2.3.1.158
2.3.1.20
Triacylglycerol
1,2-Diacyl-sn-glycerol 3-phosphate
3.1.3.4
2.7.1.107
1,2-Diacyl-3-(Glcβ1-6Glcβ1)-sn-glycerol
Lipoteichoic acid
2.7.8.20
2.4.1.-
2.4.1.157
SQD2
1,2-Diacyl-3-(6'-sulfo-α-D-quinovosyl)-sn-glycerol
Color
Enzyme
Sequences
red
ec:1.1.1.2 - alcohol dehydrogenase (NADP+)
gene_674|GeneMark.hmm|333_aa, gene_5801|GeneMark.hmm|312_aa
yellow
ec:2.3.1.158 - phospholipid:diacylglycerol acyltransferase
gene_2604|GeneMark.hmm|188_aa, gene_6532|GeneMark.hmm|505_aa
orange
ec:2.3.1.51 - 1-acylglycerol-3-phosphate O-acyltransferase
gene_176|GeneMark.hmm|429_aa, gene_6693|GeneMark.hmm|292_aa
green
ec:2.3.1.20 - diacylglycerol O-acyltransferase
gene_176|GeneMark.hmm|429_aa, gene_7213|GeneMark.hmm|521_aa, gene_8170|GeneMark.hmm|470_aa
blue
ec:2.3.1.15 - glycerol-3-phosphate O-acyltransferase
gene_886|GeneMark.hmm|748_aa, gene_2640|GeneMark.hmm|823_aa
pink
ec:1.1.1.72 - glycerol dehydrogenase (NADP+)
gene_3376|GeneMark.hmm|325_aa, gene_4577|GeneMark.hmm|326_aa
violet
ec:1.2.1.3 - aldehyde dehydrogenase (NAD+)
gene_2201|GeneMark.hmm|497_aa, gene_5247|GeneMark.hmm|502_aa, gene_5611|GeneMark.hmm|471_aa
light-red
ec:2.7.1.107 - diacylglycerol kinase
gene_5292|GeneMark.hmm|409_aa
Annotation already running

# Closing remarks



- **Structural annotation matters**

  Functional annotation depends strongly on the quality of structural annotation

  Common issues: gene fusions, gene splits, partial genes, over-predictions, wrong ORFs



- **Limits of functional annotation**

  Errors can propagate from databases => experimental validation is essential

  Annotation quality depends on the database used

  Many proteins remain *hypothetical* or *uncharacterized* (Depends largely on conventional experiments)

- **Beyond sequence similarity**

  High sequence similarity ≠ identical function

  Functional understanding requires additional context:

  - Pathway integration
  - Spatial/temporal regulation
  - Biological networks


The End
Thank you for your attention